Supplementary Information

What topology hold-out removes — and what it does not Tao Wang, College of Chemical Engineering, Nanjing Tech University, Nanjing 211816, China

> This Supporting Information accompanies the main text. Its sections are organised into nine thematic containers — **SI-A** (data, methods and reporting standard), **SI-B** (support audit and identification bounds), **SI-C** (fixed-test-set interventions), **SI-D** (protocol and aggregation robustness), **SI-E** (diagnostics and alternative analyses), **SI-F** (screening and elite subset), **SI-G** (ten targets and identity information) **SI-H** (ARC-MOF sources and granularity) and **SI-I** (six follow-up experiments and evidence limits). Tables are numbered `Table S1` … in the order they appear.

SI-A. Data, methods and reporting standard

SI-A.1 Production entry and verification boundary

The single authoritative record of inputs, producer commands and verification scope is `REPRODUCE.md`; `FIGURE_MANIFEST.md` is only an index of figure number ↔ filename. Final figures are drawn by `build_main_v6.py` (main figures, S3, S4, S7), by the source package's `pubfig` builder (S1, S2, S5, S6) and, for the table-of-contents graphic, by `build_toc_v6.py`. Historical rerun evidence applies only to the dated step lists documented in `REPRODUCE.md`. Archived outputs and current commands should therefore not be described as a fresh end-to-end rerun of the present package.

SI-A.2 Data availability

The two input databases are not redistributed. hMOF (ref. 2) is retrieved from the Materials Cloud Archive record `nsp7v-2dk91`; ARC-MOF v6 (ref. 15) from Zenodo `10.5281/zenodo.13891643`. The source package's `DATA.md` lists the retrieval commands, expected byte sizes and MD5 checksums, and records two upstream failure modes (intermittent HTTP 503, and silently truncated parallel downloads). Archived analysis outputs are stored under `outputs/`. `outputs/MD5SUMS.txt` is regenerated from that archive by `scripts/make_md5sums.py` and covers every file in it; `--check` verifies the manifest against the files. Table S7 has a separate SHA256 manifest under `outputs/baseline/`. Final figure files are stored under `../01_manuscript/figures/`. The manuscript and SI PDFs are built from their Markdown sources and the designated final figures.

The code deposit is described in main-text §9: the concept DOI `10.5281/zenodo.23415179` resolves to the latest deposited version, which is the version of record for every analysis in this manuscript (original pipeline plus the six follow-ups of SI-I and the sensitivity runs); the version DOI `10.5281/zenodo.23434268` refers to v1.3, which contains the original analyses only. Both this SI and the manuscript quote the same two identifiers, so the reader can tell which archived bytes support which claim.

SI-A.2b Model and experiment parameters

The working model is a random forest; tree counts, sample sizes and replication units differ by experiment and are not a single default shared across every analysis. The default is 120 trees; the 25,000-row multi-seed LOTO aggregation and the ten-target panel use 100 trees, and the S_g curve uses 100 to match the ten-target panel.

Experiment	Section	Rows	Features	Trees	Split	Replication unit
Protocol comparison	§3.2	100,000	426 / matched 415	120	random KFold; grouped GroupKFold (chemistry/linker/metal/topology); separate LOTO (11 folds)	single run
Controlled intervention	§3.3	50,000	426	120	fixed test set, training pool varied (arms a–f)	5 outer seeds; arm (c) 5×5 nested draws
Screening	§3.4	25,000	426	120	5 strategies, out-of-fold on a shared support set	5 seeds
Multi-seed LOTO aggregation	SI-D	25,000	426	100	LOTO per topology	5 seeds
Ten-target panel	SI-G	25,000	426	100	random vs topology hold-out	single run
S_g curve	SI-B	100,000	8 numeric + categorical one-hot (426 total)	100	LOTO per topology	single run
Singleton / entropy audit	§2.2, §3.1, §3.5, Table 3	full database	none (counting only, no model)	—	—	single pass
Geometric residual model (η^2)	§4.2	25,000	8 numeric descriptors	120	shuffled 5-fold KFold	single run
Single-group residual predictor (η^2)	§4.2	25,000	one-hot for the named grouping variable	120	shuffled 5-fold KFold	single run

The hMOF release's 42 fields group as 28 property-and-error fields, 8 numerical structure descriptors, 5 identity fields (metal centre, two linkers, functional group, topology) and 1 `MOFname` identifier. Models use the 8 descriptors plus the categorical one-hot blocks of §2.3.

SI-A.3 The 22-rule grouped-split reporting checklist

The complete checklist is reproduced here (the one-page summary table and the rules); the long-form evidence per rule lives in `REPORTING_CHECKLIST.md` alongside the runnable `scripts/confound_diagnostic.py`. The three principles of main-text §4.3 are the condensed form of these 22 rules.

#	Rule	Why it matters here
A1	State n beside every exclusivity / entropy figure	both are sample-size dependent: 98.9% at n = 25,000 vs 93.2% at n = 324,426, same data
A2	State the granularity at which "chemistry" is defined	74.7% (metal + 2 linkers) vs 93.2% (+ functional groups)
A3	Report pooled / weighted / macro / median; choose the headline by the intended deployment population	pooled alone reads +0.381 where macro reads −0.541
A4	Check target variance on both sides before subtracting R^2 across splits	a 7.7× variance gap reversed the direction of the claim
A5	State why, and whether, a ratio target is log-transformed, including zero handling	selectivity R^2 swings +0.42 … −0.06 without it

#	Rule	Why it matters here
A6	Report the number of *valid* folds	a zero-variance fold silently turns `macro` into NaN (10–11 folds here, not 11)
B1	Disclose the three diagnostic numbers in the paper, not only in the SI	single-group fraction, conditional-entropy ratio, count of pairs sharing chemistry
B2	Describe a grouped split by its held-out groups and the covariate shifts that accompany them; reserve claims about an isolated group effect for designs that identify that effect	predicting for an unseen topology is a legitimate deployment task; it is the *isolated group effect* that needs an identifying design
B3	Report a ranking metric on a *shared support set* where a screening decision is the use case	top-1% hit rate fell 50.4% → 19.3% (five seeds) across the split, on the same 25,000-sample support set
B4	Do not use random compression as a chemical-comparability correction (it may still serve as a null model or sensitivity analysis)	compression only coarsens, so it manufactures the difference it claims to remove
B5	Report n, M and the composition frequency distribution; r = n/M alone does not make libraries comparable	two libraries can differ 100× in M at similar n; matching n matches nothing
B6	Compare groupwise metrics at a common support set, or state the coverage and distribution of each population	a different support set is a different denominator, and those cannot be subtracted
B7	Stratify the diagnostic by source sub-library	needs no cross-database comparison; range here 86.1% – 100.0%
C1	Fix the test set and change only the training set for a training-intervention effect (this does not by itself isolate jointly-changing chemistry and geometry)	supports a training-intervention reading, not an automatic isolation of the chemical factor
C2	Add a training-size-matched arm	Tests whether random thinning to the same size reproduces the contrast; it does not isolate chemistry or exclude size–distribution interactions.
C3	State how many independent contrasts you actually have	one pair is not a population
C4	Removing a topology with no shared chemical keys does not directly remove matched chemical examples, but it can still change training size, descriptor coverage, and the target distribution	it is not a chemistry intervention, and it is not a pure sample-count intervention either
D1	Do not treat an ordered top-N slice as a random sample; state the sampling rule	enumerated files are sorted by topology, so the first N rows are a handful of topologies
D2	Do not feed bootstrap replicates of one sample to a between-group test	resamples are not independent observations; the p-value is pseudo-replicated
D3	Distinguish "not detected" from "zero"	median −0.011 at p = 0.28 is not proof of absence
D4	Assert on row counts and column matches before computing	an empty frame once skipped a whole check silently
D5	Verify downloads by HTTP status *and* MD5 *and* file tail	two files were truncated while `curl` still exited 0

Minimal disclosure block. If one wants a single paragraph to paste into a methods section:

> Grouped validation evaluates prediction for groups absent from training. When group identity is associated with chemical composition, this evaluation also includes chemical-support shift and does not isolate an effect attributable to group identity alone. For our data (n = N; chemistry defined as ...), **X%** of distinct compositions occur in a single `<group>`, and H(G|C)/H(G) = **Y** under the stated weighting convention. We report these diagnostics alongside the evaluation population, valid-fold counts, aggregation definitions and, where screening is the intended use, ranking performance on a shared support set.

What the checklist cannot tell you. These diagnostics do not identify a causal contribution of topology to prediction error. The nonzero conditional entropy does not by itself establish a usable matched contrast; that requires pairwise chemical overlap and sufficient observations. The diagnostic can be applied to experimentally derived libraries, but the empirical coupling measured here should not be assumed to carry over without a new audit. It is not a novelty claim: several 2026 studies report the split effect and its target-dependence; the checklist addresses the premise underneath them.

SI-A.4 Input validity: legal domains and what was done about them

Seven of the eight numeric descriptors of the hMOF release contain negative values, and the release does not document what they encode. The counts are archived per field in `outputs/input_validity_audit.csv`; in the full 324,426-structure database, **169 rows (0.052%) carry at least one negative descriptor** — surface area and void fraction 169 each, the three pore-diameter fields 138 each, volume and weight 2 each, and void volume none — and the per-field sets overlap, so the union is 169 rather than the **756** that summing the seven non-zero columns gives. No descriptor is missing; the target has no missing and no negative value, and its 1,901 zeros are retained because a zero uptake is a physically defined outcome rather than a missing record. Zero surface areas (28,528) and zero void volumes (6,467) are likewise retained: only negative values are treated as non-physical here, and that choice is a judgement, not a documented property of the release.

The affected rows are not spread evenly over topologies (acs 114, nbo 38, pcu 10, tbo 4, sra 2, the 1). Their incidence in the analysis samples is 15, 18, 17, 10 and 10 rows at n = 25,000 for seeds 42, 143, 244, 345 and 446 (0.060%, 0.072%, 0.068%, 0.040% and 0.040% of each sample), 29 at n = 50,000 and 55 at n = 100,000 for seed 42. Three of the five 25k samples therefore carry a slightly higher share than the 0.052% database rate and two a slightly lower one; the sample rates are not uniformly inflated, and none of these counts is large enough for a rate comparison to be informative on its own. Three consequences are recorded rather than assumed away.

1. **Figure S2 is refitted on valid rows.** The PDP model of the previous revision was

fitted on all 25,000 sampled rows and its quantile grid was computed over the same rows, so the void-fraction grid reached −0.468 and the surface-area grid −0.5. The grid is now computed only over rows with no negative descriptor, every grid point is asserted to lie inside the observed valid domain, the model is refitted on those rows, and the historical behaviour remains available as `prepare_figS2_pdp_data.py --keep-sentinels`. The two versions are different models, not two renderings of one; the newer one is the one shown.

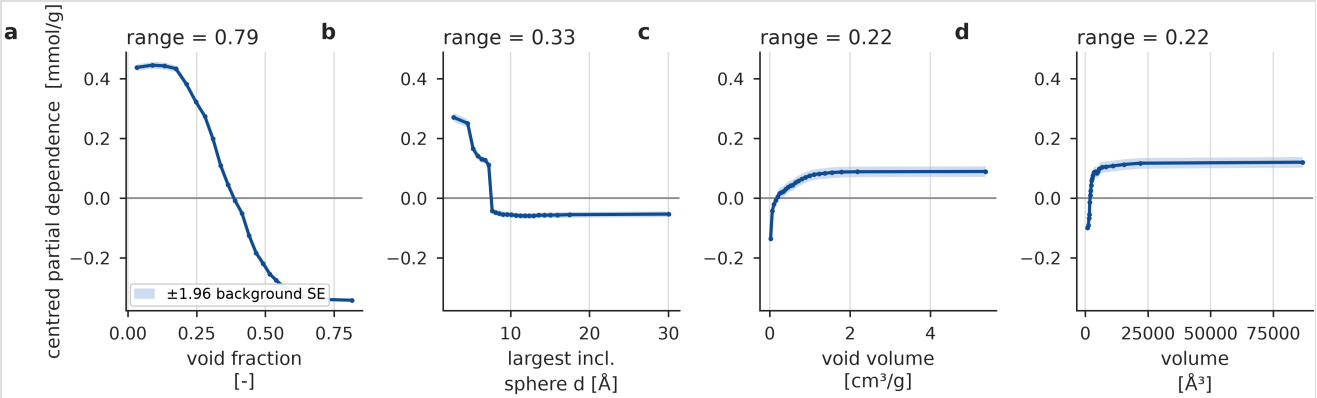


Figure S2. Centred partial dependence of the four leading structure-derived descriptors. The model is a random forest fitted to the **full analysis sample** of 24,985 rows (25,000 sampled with seed 42, minus the 15 rows that carry a negative descriptor value); it is not a training fold of the random split, and the figure footer says so. Partial dependence was calculated by replacing the selected descriptor with each grid value and averaging predictions over a fixed 4,000-row background; the grid of each descriptor is restricted to the observed valid domain, and every grid point is asserted to lie inside it (`outputs/pdp_geometry_grids.csv` records the range and the per-field invalid count). Curves are centred on the offset archived in the meta table, 0.519784 mmol/g, which is the mean of the per-tree root values (mean of tree means) rather than an explicit mean[f(background)] over the 4,000 background rows. Shading is ±1.96 background standard errors of the prediction mean, conditional on the fitted model; it is not a confidence interval for a causal effect, and a marginal substitution can produce feature combinations that lie outside the joint distribution, so the curves do not identify a causal response. PDP and SHAP describe different quantities. Descriptors, largest range first: void fraction (0.79), largest included sphere diameter (0.33), void volume (0.22), volume (0.22). *Source data:* `outputs/pdp_geometry_curves.csv` , `outputs/pdp_geometry_meta.csv` , `outputs/pdp_geometry_grids.csv` ; producer `prepare_figS2_pdp_data.py` .

1. **The model-based endpoints were re-run as a paired sensitivity (screening and

intervention, five seeds; seed 42 is detailed below).** Folds and scoring rows are held fixed, variant A trains on every row of the sample, variant B drops the non-physical rows from the *training* part of the same folds, and both are scored on the same legal rows, so the contrast is the preprocessing step and not a change of evaluation set. Variant A scored on all rows reproduces the archived seed-42 values exactly (screening: random top-1% 0.508, topology 0.132, R^2 0.7066 / 0.3660; arms: nbo a/b/e MAE 0.1104 / 0.1797 / 0.1257 and pcu 0.2403 / 0.2407 / 0.2298), which is the fidelity gate for the re-run; the gate is automated in `sensitivity_invalid_rows.py` (a mismatch above 5e-4 exits non-zero, `--gate-only` re-checks a finished table).

Screening (25,000 rows, seed 42, 15 non-physical rows). On the legal rows the random arm moves from 0.508 to 0.504 at k = 1%, from 0.5653 to 0.5709 at 5% and from 0.6717 to 0.6705 at 10%; the topology arm moves from 0.132 to **0.112** at k = 1%, from 0.3339 to 0.3299 at 5% and from 0.4235 to 0.4311 at 10%. R^2 and MAE are essentially unchanged (0.7074 → 0.7082 and 0.3674 → 0.3662; MAE 0.1589 → 0.1587 and 0.2578 → 0.2575) while the top-1% overlap moved, so the aggregate error and the top-k metric do not respond together. A stable aggregate does not identify or exclude a calibration change, and no such inference is drawn here. At this seed the random-minus-topology gap at k = 1% widens slightly, from 0.376 to 0.392; the five-seed summary below shows that this direction is not representative of the five-seed set. *Intervention (50,000 rows, seed 42, 29 non-physical rows).* nbo-side MAE moves from 0.1100 to 0.1114 (arm a), 0.1794 to 0.1774 (arm b) and 0.1252 to 0.1250 (arm e); pcu-side MAE moves from 0.2403 to 0.2429, 0.2407 to 0.2419 and 0.2298 to 0.2311. The nbo a→b absolute difference is +0.06946 mmol/g on the historical fit and +0.06602 mmol/g on the cleaned fit (+63.1% against +59.2% when scaled by the arm-a mean), and the pcu a→b change stays within ±0.5% with a sign that flips between the two variants, while the pcu a→e difference is −0.01053 mmol/g and −0.01181 mmol/g — the direction-dependent pattern and the near-zero pcu change are preserved at this seed.

Five-seed summary (42/143/244/345/446). The same paired design was run for the other four seeds. Across the five seeds, mean random top-1% recovery moves from 0.504 to 0.502 and mean topology top-1% recovery from 0.193 to 0.198; the per-seed topology change ranges from −2.0 to +2.4 pp (mean +0.6 pp), and the random-minus-topology gap change ranges from +1.6 to −2.8 pp (mean −0.8 pp). The seed-42 widening is therefore not representative: the sign of the screening change varies by seed. The intervention directions are stable: nbo a→b is positive in all five seeds (mean +0.0746 → +0.0750 mmol/g) and pcu a→e is negative in all five (mean −0.0131 → −0.0127). Per-seed values are in `outputs/sensitivity_five_seeds_summary.csv`, produced by `scripts/summarize_sensitivity.py`.

Scope: five seeds for screening and the a/b/e arms; the remaining screening strategies were **not** re-run on the cleaned sample. The affected rows are ~0.06% of each 25,000-row screening sample and ~0.06% of each 50,000-row arm sample.

LOTO cleaned sensitivity (100,000 rows, 11 folds). The Table S8 leave-one-topology-out run was repeated with the 55 non-physical rows removed from each training fold, scoring on the same legal test rows. The historical-all run reproduces Table S8 exactly; historical-legal vs cleaned-legal give pooled 0.3823 → 0.3695, weighted −0.0579 → −0.0631, macro −0.5399 → −0.4973 and median 0.2927 → 0.2814. The four-rule signs are unchanged (pooled positive, macro negative), so the sign reversal in Table 1 Panel B is not an artefact of the 55 non-physical rows. This is a single 100k run; the 25k five-seed version is reported next. *Source data:* `outputs/loto_cleaned_per_topology_100k.csv`, `outputs/loto_cleaned_aggregates_100k.csv`; producer `scripts/loto_cleaned_sensitivity.py`.

25k five-seed LOTO cleaned sensitivity. The 25k LOTO design (100 trees, five seeds) was repeated with the same paired rule. The historical-all run reproduces Table S9 exactly. Cleaned-minus-historical-legal differences are small and not systematic: pooled +0.0011 (sd 0.0102), weighted +0.0001 (0.0107), macro −0.0052 (0.0236) and median −0.0017 (0.0293). After cleaning, pooled is positive and macro negative in all five seeds; weighted is negative in four of five and flips to +0.0054 at seed 345 (historical −0.0038), i.e. the near-zero weighted sign is not stable. The aggregation-dependent sign reversal is therefore not a non-physical-row artefact. *Source data:* `outputs/loto_cleaned_per_topology_25k.csv`, `outputs/loto_cleaned_aggregates_25k.csv`; producer `scripts/loto_cleaned_25k.py`.

1. **The model-free support audit was re-run both ways.** Dropping the 169 rows leaves the

coarse composition count at 1,622 and the single-topology composition fraction unchanged at 74.7226%; the structure-weighted unseen fraction moves from 61.3434% to 61.3350% and H(T|C)/H(T) from 0.179942 to 0.180030. Both archived in `outputs/input_validity_audit.csv` under `support_audit_sensitivity`.

This bounds what can be claimed: the structural headline is insensitive to the treatment at the quoted precision, and the model-based endpoints were re-fitted on the cleaned sample for all five seeds for screening and the a/b/e arms. The screening top-1% effect is seed-dependent in sign, while the nbo a→b and pcu a→e directions are preserved. Both LOTO cleaned runs preserve the aggregation-dependent sign pattern (pooled positive, macro negative): 100k pooled 0.3823 → 0.3695 and macro −0.5399 → −0.4973; the 25k five-seed weighted sign flips at seed 345 but remains near zero.

SI-A.5 Composition-key order

The coarse key keeps `organic_linker1` and `organic_linker2` in file order. Sorting the pair before concatenation collapses six ordered keys into three (three pairs are reversed duplicates) and reduces the composition count from 1,622 to 1,619; the single-topology composition fraction moves from 0.7472256 to 0.7467573 (both 74.7% at the quoted precision), the structure-weighted unseen fraction is unchanged at 0.6134342, and the same seven topologies are complete hold-outs. The ordered definition is retained because it is the file's own convention and the release documentation does not say whether `organic_linker1` and `organic_linker2` encode distinct construction sites. We therefore report the unordered sensitivity instead of asserting a site semantics, and the sensitivity shows only that the support counts do not depend on the convention, not that the grouping or intervention models are invariant to it.

SI-B. Support audit and identification bounds

SI-B.1 Full per-topology decomposition

The main text reports 7 of 11 topologies as exact composite-key hold-outs and the pcu / nbo unseen fractions (60.1% / 12.3%). The complete eleven-row decomposition, at the coarse granularity on the full 324,426-structure database, is:

Table S1. Per-topology chemical-support loss (coarse key, full hMOF database).

Held-out topology	n_samples	seen fraction	unseen fraction
pcu	205,361	0.399	**0.601**
nbo	49,464	0.877	**0.123**
sra	23,962	0.000	1.000
etb	19,682	0.000	1.000
acs	13,655	0.000	1.000
bcu	8,795	0.000	1.000
pts	1,522	0.000	1.000
fof	1,256	0.000	1.000
tbo	363	0.110	0.890
the	352	0.000	1.000
rht	14	1.000	0.000

Source data: `outputs/chemistry_seen_by_topology.csv`. "Completely unseen" holds only for the composite chemistry key, not for each building block separately — a topology can have no composite-key counterpart while its metal centre or a linker still recurs in training.

Component breakdown (same counting rules). To show which part of the composite key drives the exclusivity, the same full-database count was repeated with three key definitions: metal centre only (M = 7), the ordered linker pair (M = 602), and the conjunction of metal and both linkers (M = 1,622). The single-topology composition fractions are 57.1%, 46.0% and 74.7%; the structure-weighted unseen fractions are 28.7%, 23.1% and 61.3%; and H(T|C)/H(T) is 0.345, 0.512 and 0.180. Neither component alone reproduces the conjunction: the metal key is too coarse (7 values), and the linker pair alone leaves more than half of the compositions multi-topology. The conjunction is therefore the right headline, while the component rows show that the exclusivity is a property of the combination rather than of one building block. *Source data:* `outputs/component_breakdown.csv`, produced by `scripts/component_breakdown.py` from the same raw hMOF release. The three key fields used here (`metal_linker`, `organic_linker1`, `organic_linker2`) are complete in this release: 0 missing and 0 literal `NA` values in all 324,426 rows, so the tuple-based grouping has no missing-value category; the script fails closed if a future input contains one. The functional-group field is not part of this breakdown; its 1,997 unannotated rows are treated as one fine-granularity value in SI-B.2, as stated in §2.1.

SI-B.2 Entropy, singletons and the two directions

The structure-weighted ratio is H(T|C)/H(T) = 0.180 (coarse), so conditioning on coarse chemistry removes 82.0% of the entropy in the topology label; the reverse ratio H(C|T)/H(C) = 0.860 says topology removes only 14% of composition uncertainty (H(C) = 10.438 bit, H(T) = 1.777 bit, H(C|T) = 8.981 bit). The fine-granularity ratio is 7.8%, but the fine space is dominated by singletons: at least 2M − N = 80,536 of 202,481 fine compositions occur exactly once, and the non-singleton single-topology fraction is 79.1% among the 66,172 non-singleton compositions. The first-order Miller–Madow correction is $\Delta H(T|C) = (K_{TC} - K_C)/(2\,N \ln 2)$: 0.000912 bit (0.3%) at coarse and 0.0307 bit (22%) at fine, against H(T|C) = 0.320 and 0.139 bit. The coarse headline is barely affected by the observed singleton count (21 of 1,622, i.e. 1.3%; 74.4% among non-singletons against 74.7% overall). The correction is a first-order diagnostic, not a proof of unbiasedness. The `NA` fill convention inherited from `chemistry_seen.py` does not merge distinct values here: the three coarse key fields have 0 missing and 0 literal `NA` values, and the fine key's only missing category is the 1,997 unannotated functional-group entries (no literal `NA`); this was checked on the full release. *Source data:* `outputs/singleton_audit.csv` (both entropy ratios, the singleton counts, the non-singleton fractions and all four Miller–Madow corrections); the coarse/fine marginals are also in `outputs/confound_full.csv`.

SI-B.3 Pairwise overlap and the S_g negative result

Table S2. Compositions spanning more than one topology, at both granularities (full hMOF database, 324,426 structures). Each row's last three columns partition the multi-topology set exactly.

Granularity	Distinct compositions M	Multi-topology compositions	Share of M	From pcu ↔ nbo	From rht ↔ tbo	From the other 53 pairs
Coarse (metal + 2 linkers)	1,622	**410**	**25.3%**	**408** (99.5% of the multi-topology set)	2	**0**
Fine (+ functional groups)	202,481	**13,825**	**6.8%**	**13,821**	4	**0**

Source data: `outputs/confound_full.csv` (the two M values and the composition-level single-topology fractions) and `outputs/jaccard_full.csv` (the 55-pair table). No hMOF composition spans more than two topologies, so the multi-topology set is exhausted by the two named pairs at each granularity and the arithmetic closes exactly (408 + 2 = 410; 13,821 + 4 = 13,825). The fine-granularity counts keep the `fillna` convention of main-text §2.1 (a missing functional group is one additional value, not a dropped row) and are inflated by singleton compositions, so they are a robustness check only.

The S_g analysis asks whether the unseen-chemistry fraction predicts the hold-out cost of the same topology, measured on one common run (100,000 samples, the 100-tree, seed-42 model, leave-one-topology-out; `scripts/s_g_curve.py`, `outputs/s_g_curve.csv`). That run is distinct from the 120-tree LOTO run behind Table S8, which is why the two tables' per-topology R^2 values differ in the second decimal (for example etb −2.55 here against −2.49 there); neither is a correction of the other. This exploratory analysis did not establish a statistical association. Seven of the eleven topologies share S_miss = 1.000 yet their pooled R^2 spans −2.55 (etb) to +0.47 (bcu). Across the ten topologies with n_test ≥ 100, Pearson r = +0.10 (p = 0.78) and Spearman ρ = +0.28 (p = 0.44); excluding pcu gives r = +0.53 (p = 0.11). These small-sample results do not establish absence of an association (Figure S7).

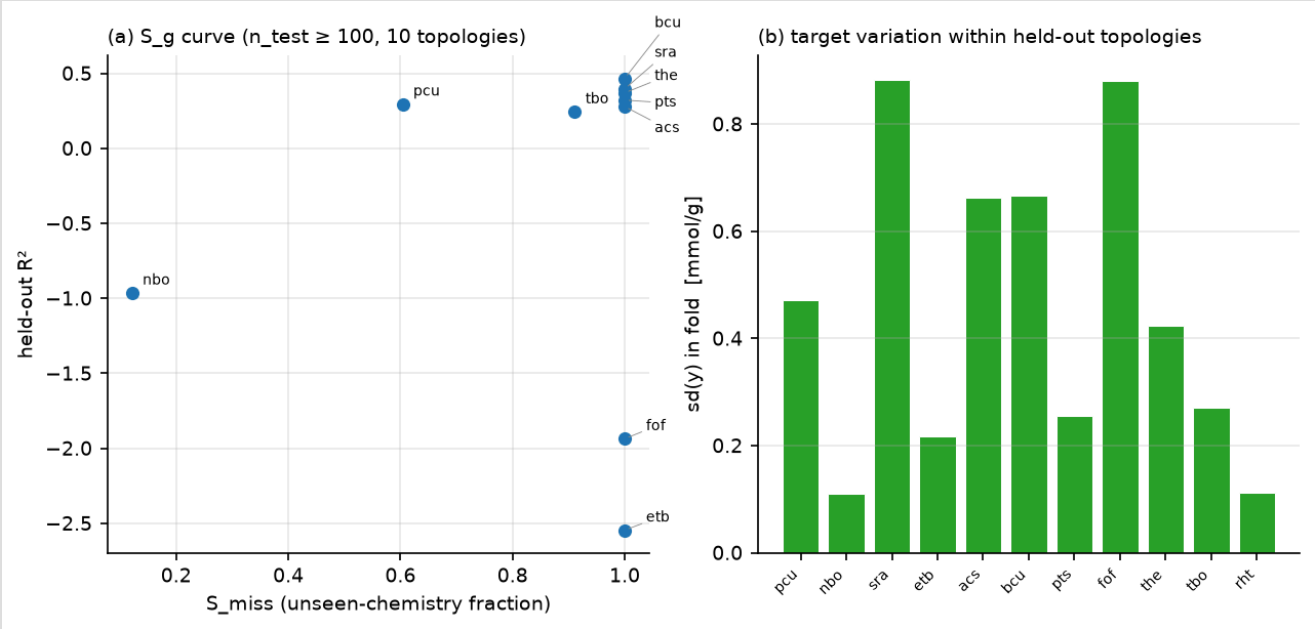


Figure S7. The S_g curve: does a topology's unseen-chemistry rate order its hold-out cost? (a) Each held-out topology is one point: horizontal = S_miss (fraction of that topology's structures with no chemical counterpart in training), vertical = that fold's held-out R^2; panel (a) is restricted to the ten topologies with n_test ≥ 100, while panel (b) shows all eleven. Seven points sit at S_miss = 1.000 and still span R^2 from −2.55 (etb) to +0.47 (bcu); the exploratory association was not statistically established (Pearson r = +0.10, p = 0.78; Spearman ρ = +0.28, p = 0.44). (b) Target variation within held-out topologies (sd(y) from 0.11 to 0.88 mmol/g); the pcu fold accounts for 63.4% of the scored test rows. *Source data:* `outputs/s_g_curve.csv`.

Table S3. S_g curve: per held-out topology, on one common run (100,000 samples).

Held-out topology	n_test	S_miss	r = n_train/M_train	sd(y) in fold	pooled R^2
rht	5	0.000	62.9	0.113	−3.02
nbo	15,200	0.121	54.8	0.110	−0.97
pcu	63,447	0.604	38.8	0.472	+0.29
tbo	101	0.911	63.7	0.270	+0.24

Held-out topology	n_test	S_miss	r = n_train/M_train	sd(y) in fold	pooled R^2
acs	4,120	1.000	63.0	0.663	+0.28
bcu	2,707	1.000	64.1	0.667	+0.47
sra	7,323	1.000	65.7	0.883	+0.40
etb	6,091	1.000	64.6	0.216	−2.55
fof	412	1.000	63.1	0.880	−1.94
pts	487	1.000	62.9	0.254	+0.32
the	107	1.000	63.1	0.424	+0.37

Source data: `outputs/s_g_curve.csv`. The per-fold R^2 values sit on different test populations, so their differences cannot be attributed uniquely to the model or protocol; the panel is a descriptive sensitivity analysis. R^2 on the *same* test rows is a strong control (the denominator does not change), and R^2 on different populations can still be reported — it just cannot be differenced as a model or protocol effect. The exploratory S_miss–R^2 analysis does not establish an error ordering across topology groups. It uses a different outcome and model configuration from the absolute-MAE associations in SI-I.4; neither analysis calibrates a prospective risk rule. The fixed-test intervention in §3.3 addresses a separate within-population training-pool contrast.

SI-C. Fixed-test-set interventions

SI-C.1 Arms and the two size-matched designs

The six prespecified arms (a–f) are defined in the main text Methods, which also defines the post hoc arm (g) used only as a control in SI-C.4. Two distinct size-matched (c) designs exist and are kept separate: the **nested (c)** (0.1071 ± 0.0048 MAE on nbo, 0.2364 ± 0.0031 on pcu, from `scripts/diversity_control.py`, `outputs/composition_matched_control.csv`) makes five size-matched draws **within each of the five outer seeds** (25 draws), with the SD taken across the five outer seed means, never across the 25 draws; the **seed-42 repeated-draw (c)** (0.1030 ± 0.0031, from `scripts/confound_diagnosis.py --size-match`, `outputs/isoreticular_control_sizematched.csv`) uses five draws at the seed-42 baseline. These are different replication designs, not two precisions of one estimate.

SI-C.2 Arm (d) construction limitation

Arm (d) reproduces (b)'s topology/composition profile, which confines it to (b)'s own row set (42,329 − 31,682 = 10,647); it is therefore an internal-consistency check, not an independent mechanism contrast. Its per-seed composition counts (496 compositions against (c)'s 1,484) show that composition count alone does not order the result.

SI-C.3 Per-seed tables

Table S4. Controlled contrast at seed 42: fixed test set, changing training set.

Test topology	Training set	n_train	R^2	MAE	RMSE	MAE/sd
nbo (n = 7,671)	with the partner pcu	42,329	−0.875	**0.1104**	**0.1456**	1.038
nbo (n = 7,671)	**pcu removed**	10,647	−4.383	**0.1797**	**0.2467**	1.690

Test topology	Training set	n_train	R²	MAE	RMSE	MAE/sd
pcu (n = 31,682)	with the partner nbo	18,318	+0.268	0.2403	0.4038	0.509
pcu (n = 31,682)	**nbo removed**	10,647	+0.262	0.2407	0.4055	0.510

Table S5. Arms (e) and (f) at seed 42: partner-shared-composition removal and size-matched control. Every cell is a seed-42 value; `n_shared_removed` for arm (e) is the number of partner rows this seed actually removes (12,300 at seed 42 on the nbo side and 6,765 on the pcu side; the five-seed means are 12,342 and 6,689, the numbers quoted in main-text §3.3).

Test topology	Training set	n_train	MAE	RMSE	R²	n_shared_removed	n_compositions
nbo	(a) with partner	42,329	**0.1104**	0.1456	−0.875	—	1,530
nbo	(b) partner removed	10,647	**0.1797**	0.2467	−4.383	31,682	493
nbo	(e) shared-composition rows removed	30,029	**0.1257**	0.1660	−1.438	12,300	1,143
nbo	(f) size-matched from (e)	10,647	**0.1211**	0.1667	−1.459	—	1,117
pcu	(a) with partner	18,318	0.2403	0.4038	+0.268	—	923
pcu	(b) nbo removed	10,647	0.2407	0.4055	+0.262	7,671	493
pcu	(e) shared-composition rows removed	11,553	**0.2298**	0.3949	+0.300	6,765	536
pcu	(f) size-matched from (e)	10,647	**0.2322**	0.4005	+0.280	—	535

Table S6. Full per-seed MAE [mmol/g] for the five outer seeds and every arm, nbo and pcu test sets. Arm (c) is the mean of that seed's five size-matched inner draws.

Test	seed	a	b	c (nested)	e	f
nbo	42	0.1104	0.1797	0.1103	0.1257	0.1211
nbo	143	0.1030	0.1628	0.1031	0.1279	0.1325
nbo	244	0.1077	0.2013	0.1116	0.1453	0.1317
nbo	345	0.1082	0.1809	0.1096	0.1364	0.1244
nbo	446	0.1034	0.1807	0.1009	0.1275	0.1152
pcu	42	0.2403	0.2407	0.2374	0.2298	0.2322
pcu	143	0.2465	0.2383	0.2369	0.2318	0.2262
pcu	244	0.2387	0.2387	0.2395	0.2287	0.2308
pcu	345	0.2399	0.2401	0.2370	0.2262	0.2259
pcu	446	0.2466	0.2401	0.2311	0.2299	0.2368

Arm (d) uses the same row set as (b) and is therefore not tabulated separately. The absolute paired 95% interval for the (a)→(b) change is a t-interval on the five within-seed differences; the percentage version divides that difference by the with-partner mean, with the denominator treated as fixed.

Table S7. Training-mean-predictor baseline, MAE divided by test-set SD (five-seed mean ± SD, generated from the raw uptake by the archived script under the original sampling rule). Each arm's baseline is its own training-fold mean, so it differs between arms. The `MAE_over_sd` column of the archived `isoreticular_control_seeds.csv` is the random forest's standardised error, **not** this baseline.

Test topology	arm	training-mean predictor MAE / test-set SD
nbo	a (with partner)	3.438 ± 0.085
nbo	b (partner removed)	5.171 ± 0.143
pcu	a (with partner)	0.702 ± 0.004
pcu	b (partner removed)	0.969 ± 0.009

Both arms beat their own baseline in all five seeds on both test topologies. For each seed and arm, the constant prediction is the mean of that arm's training uptake. MAE is divided by the test-target SD (ddof=0); the table reports the mean and sample SD (ddof=1) of the five seed-level ratios. No model is fitted for this calculation. *Source data:* `outputs/baseline/training_mean_baseline_seeds.csv` and `outputs/baseline/training_mean_baseline_summary.csv`. *Producer:* `scripts/training_mean_baseline.py`; input checksums, execution timeline and verification commands are recorded in `REPRODUCE.md`.

The five-seed means of every arm (the headline numbers of main-text Table 2) are: nbo MAE — a 0.1065 ± 0.0032, b 0.1811 ± 0.0136, c 0.1071 ± 0.0048, e 0.1326 ± 0.0082, f 0.1250 ± 0.0073; nbo RMSE — a 0.1352 ± 0.0077, b 0.2405 ± 0.0136, c 0.1375 ± 0.0109, e 0.1745 ± 0.0127, f 0.1697 ± 0.0145; pcu MAE — a 0.2424 ± 0.0038, b 0.2396 ± 0.0010, c 0.2364 ± 0.0031, e 0.2293 ± 0.0020, f 0.2304 ± 0.0045; pcu RMSE — a 0.4030 ± 0.0021, b 0.4024 ± 0.0030, c 0.4002 ± 0.0044, e 0.3935 ± 0.0032, f 0.3961 ± 0.0066. The (c) SD is the nested value (SD across the five outer seed means of five inner draws each), not the flat SD over the 25 draws. *Source data:* `outputs/isoreticular_control_seeds.csv` (a, b), `outputs/composition_matched_control.csv` (c, d), `outputs/arm_e_partner_coverage.csv` (e, f). Arm (f)'s n_train equals arm (b)'s **by construction** — it draws that many rows from arm (e)'s pool — so the two agree seed by seed, and the script asserts it.

SI-C.4 A post hoc random-partner-row control

The a–f design does not separate “remove the shared-composition partner rows” from “remove partner rows in general”. We therefore added a post hoc arm (g): from arm (a), remove at random the same number of partner rows that arm (e) removes, without conditioning on whether their composition occurs in the test set. This control was run after the a–f design was fixed; it is a post hoc check, not a new preregistered arm, and it does not change Table 2.

The original five-seed means are a/e/g = 0.2424/0.2293/0.2341 mmol/g on pcu and 0.1065/0.1326/0.1189 on nbo. A paired-seed reanalysis gives e−g = −0.004850 mmol/g for pcu (95% t interval [−0.010086, +0.000385]) and +0.013668 for nbo ([+0.009000, +0.018337]). The pcu interval includes zero. Thus e's improvement over a does not establish a composition-specific advantage over same-count random partner deletion. These are contrasts between training interventions within one chemistry-overlapping pair; they do not isolate shared composition from descriptor or target-distribution changes. The five sample redraws overlap within the same database, so the intervals do not represent independent external validation. A second adequately supported chemistry-matched pair is not available in hMOF (rht ↔ tbo has only two shared compositions). *Source data:* `outputs/partner_random_g.csv`, `outputs/arm_e_partner_coverage.csv` and `outputs/followup_20261023/experiment5/main_a_e_g_paired.csv`. The fixed-size matched replacement follow-up (SI-I.5) addresses a different intervention and retains residual composition and outcome differences.

SI-D. Protocol and aggregation robustness

SI-D.1 Aggregation rules and valid folds

The four rules (pooled / weighted / macro / median) and their LOTO values (+0.3814 / −0.0589 / −0.5414 / +0.2782) are in the main text. All eleven fold R^2 values are defined in the 100,000-row LOTO run. In the 25,000-row runs, seeds 42, 345 and 446 each contain a singleton rht test fold, giving ten valid fold R^2 values; seeds 143 and 244 have eleven. Pooled scoring retains all predictions, whereas macro and median aggregation use defined fold scores. Across the five 25k seeds, pooled's dispersion is 4.2% of its mean while macro's is 91.5% (Figure S4; the four rules of one run and their eleven folds are in Figure S5).

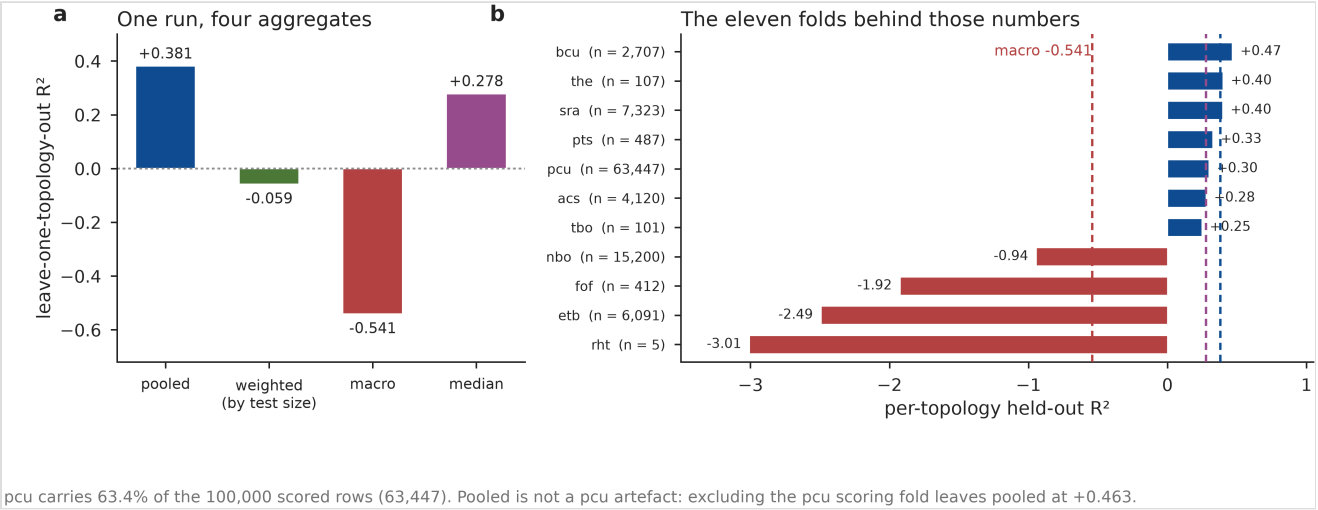


Figure S5. Leave-one-topology-out R^2 for each topology under four aggregation rules. (a) The four aggregates of one run (pooled +0.381, weighted −0.059, macro −0.541, median +0.278). (b) The eleven folds behind them, ordered by R^2, with the pooled, macro and median values marked by dashed rules (weighted sits near zero and is shown only in panel a); a fold's test-set size is printed with its name. Excluding pcu, weighted is −0.678 and macro −0.625; pooled without the pcu scoring fold is +0.4629. *Source data:* `outputs/metrics_audit.csv`, `outputs/loto_per_topology_final.csv`.

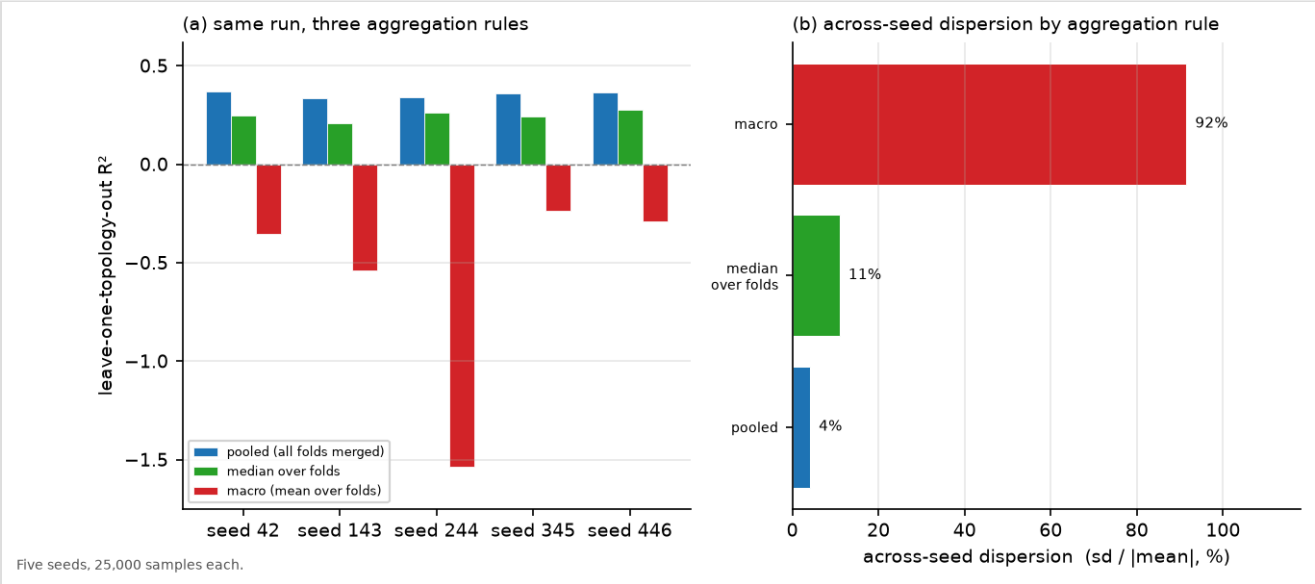


Figure S4. Across-seed dispersion by aggregation rule. (a) Pooled, median and macro on the same run with the zero line. (b) The across-seed dispersion of each rule (sd / |mean|): pooled 4.15%, median 11.03%, macro 91.53%. The figure reports the three rules' dispersion; it does not attribute the difference to any single cause. *Source data:* `outputs/multi_seed_uptake25k_fix.csv`.

"Excluding pcu" is a scoring-fold exclusion, not a retraining. Predictions from the other ten test folds were retained without refitting; their original training pools were unchanged. Recomputing pooled over those ten folds gives +0.4629 over the remaining 36,553 test rows; the per-group aggregates excluding pcu (weighted −0.678, macro −0.625) are different summaries of the same remaining folds. *Source data:* `outputs/metrics_audit.csv`.

Table S8. Leave-one-topology-out per held-out topology on the 100,000-sample single run (120 trees, `scripts/loto_full_metrics.py recompute`). Every fold has a defined R^2 (11/11 valid: n_test ≥ 2 and SST > 0). MAE and RMSE are in mmol/g; R^2 is normalised within each fold by that fold's own target sum of squares. The four aggregates of this run are pooled +0.381, weighted −0.059, macro −0.541 and median +0.278 (`outputs/loto_aggregates_100k.csv`).

Held-out topology	n_test	MAE	RMSE	per-topology R^2
pcu	63,447	0.2382	0.3952	+0.298
nbo	15,200	0.1153	0.1537	−0.943
sra	7,323	0.4793	0.6862	+0.396
etb	6,091	0.3125	0.4041	−2.492
acs	4,120	0.4193	0.5633	+0.278
bcu	2,707	0.3056	0.4866	+0.467
pts	487	0.1153	0.2087	+0.326
fof	412	1.2610	1.5029	−1.924
the	107	0.1660	0.3275	+0.399
tbo	101	0.1556	0.2330	+0.247
rht	5	0.1876	0.2018	−3.007

MAE and R^2 need not rank topologies identically, because R^2 normalises squared error by each fold's target variation. For example, nbo has a lower MAE than pcu (0.115 versus 0.238 mmol/g), but a lower R^2 (−0.943 versus +0.298). These metrics therefore provide complementary descriptions of predictive performance across different test populations.

Table S9. Leave-one-topology-out aggregation on the 25,000-sample five-seed design (100 trees; `scripts/loto_full_metrics.py recompute25k`; `outputs/loto_aggregates_25k.csv`). `weighted` is the sample-size-weighted mean of the valid per-fold R^2, $\Sigma(n_i R^2_i)/\Sigma n_i$, normalised over valid folds only; "covered" is the number of test rows those valid folds contain. Seeds 42, 345 and 446 each contain a singleton rht fold (n_test = 1), which has no defined R^2 and is excluded from weighted/macro/median but retained in the pooled score.

seed	n	pooled	weighted	macro	median	valid folds	covered
42	25,000	+0.367	−0.052	−0.354	+0.247	10	24,999
143	25,000	+0.337	−0.033	−0.538	+0.206	11	25,000
244	25,000	+0.339	−0.032	−1.539	+0.264	11	25,000
345	25,000	+0.358	−0.004	−0.238	+0.240	10	24,999
446	25,000	+0.367	−0.031	−0.289	+0.278	10	24,999

Table S10. Within- versus cross-topology R^2 per topology (25,000 samples, eight structure-derived descriptors; `outputs/within_vs_cross_final.csv`).

Topology	n	within R^2	cross R^2
bcu	705	0.810	0.496
sra	1,859	0.711	0.460
acs	1,049	0.662	0.159
pcu	15,833	0.648	0.279
nbo	3,798	0.612	**−0.312**
etb	1,481	0.477	**−4.269**

Table S11. Macro and median aggregation for the within/cross comparison (25,000 samples).

Rule	within	cross
macro	**+0.654**	**−0.531**
median	+0.655	+0.219

Median and macro differ by 0.75 because etb (−4.27) and nbo (−0.31) pull the mean; every topology moves in the same direction (within > cross for all six groups), so the disagreement is about magnitude only (Figure S3). This is a different statistic from the LOTO aggregate of §3.2 (different test folds, feature set and sample size), so the two sections' macro values (−0.541 and −0.531) are close by coincidence, not identity.

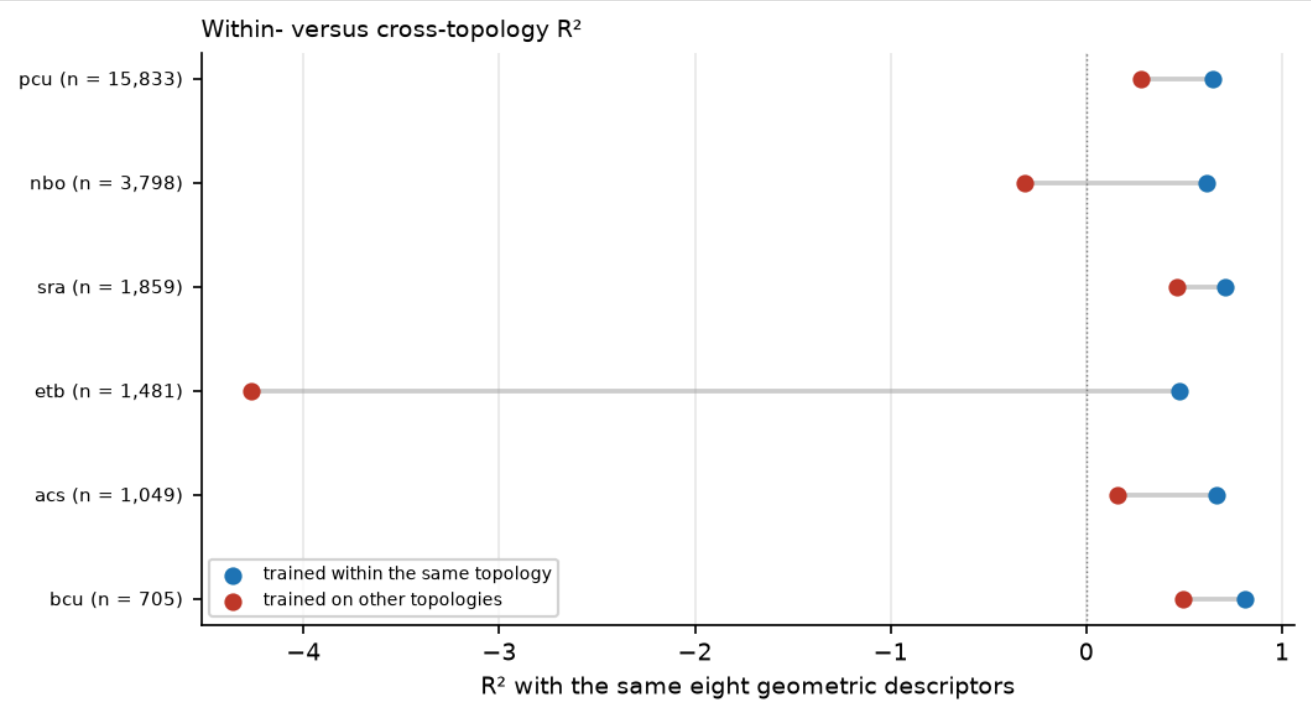


Figure S3. Within-topology against cross-topology R^2 per topology for the eight structure-derived descriptors (25,000 samples). Each row is one topology, ordered by test-set size; the connector is the within-to-cross gap and a topology's n is printed with its label. Median within 0.655 against median cross 0.219; cross-topology R^2 is negative for 2 of the 6 topologies (etb −4.27, nbo −0.31). *Source data:* `outputs/within_vs_cross_final.csv`.

SI-D.2 Model families and the second target

The random-to-topology gap was refit under gradient boosting and ridge (single fits, no tuning), confirming it is not specific to random forests; the per-family rows are in `outputs/leakage_results_gb100k.csv` and `outputs/leakage_results_ridge100k.csv`. The second target (log-selectivity) is in `outputs/leakage_results_logsel100k.csv`; the main text states the target dependence in one sentence.

SI-D.3 The 415-feature matched check

Removing the topology block from both arms (415 features) gives the chemistry-key decrease as 21.5% of the topology decrease (21.1% unmatched); the matched decrease is 0.3603 (0.3603271, rounded). This is a descriptive protocol ratio, not a causal share. *Source data:* `outputs/leakage_results_notopo100k.csv`.

SI-E. Diagnostics and alternative analyses

SI-E.1 Candidate explanations

Table S12. R^2 by nearest-neighbour distance bin (25,000 samples per scheme). The 12 bins are quantiles of the distance distribution pooled over the two schemes, with open-ended first and last bins; `n` is the number of test rows each scheme contributes to that bin. The two schemes do not hold the same rows even where the edges coincide, so each row is a pair of overlapping but non-identical populations. Producer: `leakage_mechanism.py`; the bin edges and both sample counts are fields of `fig7_nn_curve_final.csv`, not a separate rule (REPRODUCE, "Distance-bin diagnostic").

nearest-neighbour distance bin	n (random)	n (topology)	random R^2	LOTO R^2
`(-inf, 0.082]`	3,671	499	0.693	0.258
`(0.082, 0.105]`	3,435	730	0.652	0.222
`(0.105, 0.127]`	3,193	972	0.626	0.157
`(0.127, 0.149]`	2,960	1,209	0.695	0.170
`(0.149, 0.173]`	2,724	1,440	0.713	0.276
`(0.173, 0.2]`	2,320	1,847	0.670	0.292
`(0.2, 0.232]`	1,838	2,329	0.695	0.258
`(0.232, 0.271]`	1,461	2,705	0.680	0.281
`(0.271, 0.318]`	1,106	3,061	0.637	0.292
`(0.318, 0.386]`	898	3,268	0.515	0.254
`(0.386, 0.507]`	744	3,423	0.705	0.003
`(0.507, inf]`	650	3,517	0.541	**−0.775**

The two curves do not coincide in any bin, but **neither column is monotonic**, and the degradation is concentrated in the largest-distance bin (−0.775). The 25,000 rows of each scheme fall entirely into the 12 bins (3,671 + … + 650 = 25,000 for random, 499 + … + 3,517 = 25,000 for the held-out-topology scheme), so no bin was dropped by a minimum-count rule and the table preserves the full support set. Within-bin median distances differ by scheme (0.066 for random against 0.070 for the held-out scheme in the first bin), which is why the rows are keyed by bin edges rather than by a shared median. Distance-stratified and family-seen diagnostics describe differences across evaluation subsets; they do not estimate the fraction of the topology-hold-out penalty attributable to distance or chemistry, because the subsets have different target distributions and R^2 denominators. The family-level diagnostic is exploratory in the same way: the composite family (metal + both linkers + functional groups, **excluding topology**) is present in training for only 8.9% of random-split test samples, so the family-seen contrast is a weak negative rather than a clean comparison. SHAP family attribution is essentially unchanged between splits (structure-derived 82.3% → 85.1%, identity 17.7% → 14.9%) (Figure S1; partial dependence of the four leading descriptors is in Figure S2). None of these diagnostics fully accounts for the gap, and none is an orthogonal intervention, so they are descriptive rather than decisive.

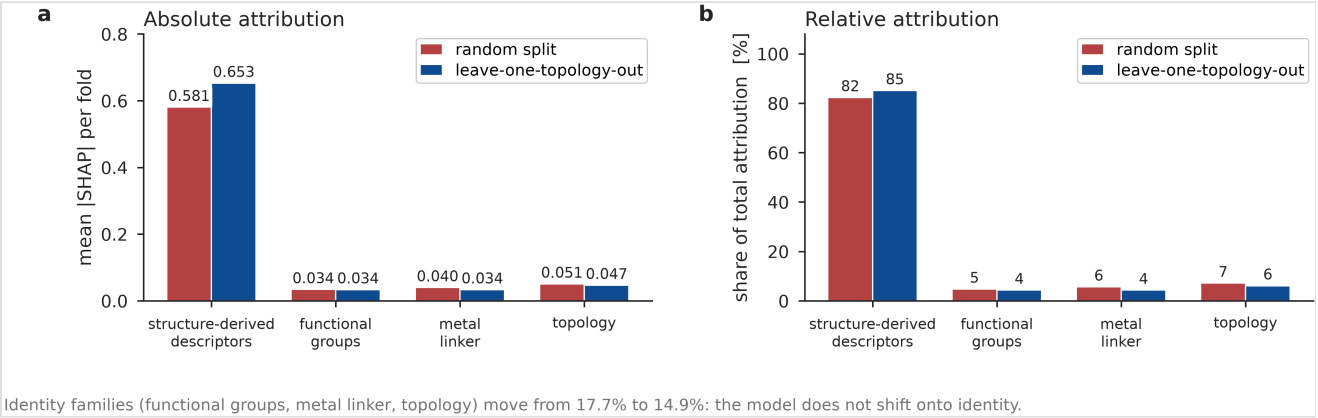


Figure S1. SHAP attribution aggregated by feature family under a random split and under topology hold-out. The attribution is dominated by structure-derived scalar descriptors in both regimes (82.3% / 85.1%). The original model assigns 17.7% and 14.9% of total mean absolute SHAP attribution to its encoded identity block. These shares do not measure the information content of chemistry or the benefit of adding linker identities (SI-I.1). Left panel: mean |SHAP| per fold; right panel: the same numbers as a share of total attribution. n_folds = 5 (random) and 11 (topology hold-out). *Source data:* `outputs/fig5_shap_family_final.csv`.

SI-E.2 The withdrawn deep-learning comparison

Table S13. Why the deep-learning comparison is not interpretable: two test sets 7.7× apart in target variance.

	Random split	**Topology hold-out (test set all pcu)**
n	500	317
Test-set target variance Var(y)	**0.5460**	**0.0706**
R^2	+0.205	**−0.448**
MAE	**0.3958**	**0.2617**
RMSE	**0.6587**	**0.3199**

The variances differ 7.7× (standard deviations 2.78×); R^2 is normalised by the test-set variance, and the hold-out side's MAE and RMSE are **smaller** on its own test population. Lower absolute error on a different test population does not establish model superiority, so these six numbers are reported only to show why the comparison is not interpretable. The R^2 and test-variance values are in `outputs/metrics_audit.csv` (the CGCNN rows; for these rows the `macro` column stores the test-set variance); the MAE and RMSE values are reported from the run log of the withdrawn arm, which is not archived. No claim elsewhere in the paper rests on these numbers. A crystal-graph arm would be interpretable only if (i) the two test sets are matched on topology composition and target variance, (ii) MAE/RMSE, not R^2, are the primary metrics, and (iii) both arms are trained to convergence; none of the three holds.

SI-E.3 Mitigation and omitted-variable analyses

Table S14. Mitigation strategies compared under a fixed protocol (25,000 samples).

Feature set	# features	Random R^2	Grouped-by-topology R^2	gap
structure-derived + building-block one-hot (baseline)	426	0.7066	0.3660	**0.341**
structure-derived descriptors only (identity removed)	8	0.6719	0.3740	0.298
structure-derived + hash encoding (32-dim)	40	0.6652	0.3762	0.289
structure-derived + **in-fold target encoding**	11	0.6998	0.2885	**0.411**

In this single-run comparison using the original identity block, removing that block changes topology-grouped R^2 from 0.366 to 0.374 while lowering random-split R^2. This does not characterize the effect of adding linker identities: the follow-up I−B comparisons use a different identity block, encoding and feature-subsampling setting (SI-I.1). Under this implementation, in-fold target encoding increased the random-versus-grouped performance gap (0.411 against the 0.341 baseline); the implementation encodes each training fold with its own whole-fold category means and no inner cross-fitting, and a canonical inner-cross-fitted implementation is not evaluated here. Hash encoding and identity removal shrink the gap by lowering the random-split R^2 far more than the grouped-split R^2 moves, so both terms of every gap are reported. *Source data:* `outputs/mitigation_results_mit25k.csv`.

SI-F. Screening and the elite subset

SI-F.1 All seeds and k values

The five-seed top-1% overlap is in the main text (50.4% → 19.3%, complete ordering 4/5). The per-seed rows, all five strategies, and the single seed-42 run (which sits at the low-recovery end, 0.132) are in `outputs/screening_utility_seed{42,143,244,345,446}.csv` and `outputs/screening_utility.csv`. The oracle shortfall is 46.3% by the ratio-of-means convention (1 − mean(selected)/mean(oracle) = 46.324993%); the per-seed mean of ratios (mean of 1 − selected/oracle) is 46.300683% — a distinct statistic that happens to round to the same 46.3%. The selected mean is 1.742 mmol/g and the oracle mean 3.246 mmol/g; the oracle differs across seeds (3.207–3.319 mmol/g, mean ± SD 3.246 ± 0.043) because each seed redraws the sample, while within a seed the oracle is the same for every strategy.

Table S15. Five-seed top-k overlap at three budgets (% of the true top-k recovered, mean ± SD across the five seeds; the same five seeds for every strategy).

Budget k	random overlap	topology overlap
1%	50.40 ± 2.48	19.28 ± 7.06
5%	56.82 ± 1.09	33.31 ± 2.06
10%	67.23 ± 0.49	40.66 ± 1.70

The gap persists at all three budgets and is not monotonic in k: 31.1 percentage points at 1%, 23.5 at 5% and 26.6 at 10% (random minus topology, five-seed means). The main text quotes the k = 1% headline and refers the 5%/10% values here rather than mixing them with the seed-42 numbers. For the k = 1% headline the five paired within-seed differences give 31.12 ± 5.26 percentage points (95% paired t interval 24.59 to 37.65; $t(4) = 13.24$, $p = 1.9 \times 10^{-4}$), computed by `scripts/screening_paired_tests.py` and archived in `outputs/paired_tests_20261024.csv`; the 5% and 10% gaps are reported descriptively only, because k = 1% is the prespecified headline budget and no test is attached to the wider budgets. As in the main text, the interval describes seed-to-seed variation within one database. *Source data:* `outputs/screening_utility_seed{42,143,244,345,446}.csv`.

Tie audit at the truncation. Row-level predictions are archived for the random and topology strategies (five seeds, two row policies: historical and cleaned). At k = 1%, 5% and 10% no predicted score tie straddles the cut in any of the 60 combinations; the maximum number of rows sharing the boundary score is one. Re-ranking the tied block by lowest source row ID instead of archived row order selects exactly the same members in every case (`outputs/finish_20261023/tie_audit.csv`, produced by `scripts/analyze_finish_20261023.py`). The chemistry-key, linker-pair and metal strategies are archived only as summary metrics, so the same audit cannot be repeated for them; the rule in the main text applies to all five but is only verified for two. For the fixed candidate pool, ties are resolved by source-row ID and the archived file stores only the selected materials, not the full score vector (SI-I.6), so no tie audit is possible there.

Candidate resampling versus model redraws. A within-sample row bootstrap (200 resamples per seed and strategy, k = 1%) gives an overlap SD of 2.1–3.0 percentage points, against a five-seed SD of 2.5 points for random and 7.1 points for topology. The two objects answer different questions: the bootstrap varies which 25,000 materials are in the candidate set under one fitted protocol, while the seed SD varies the model redraw and its training memberships. Neither estimates variation across independent MOF libraries, and the smaller bootstrap SD must not be substituted for the seed SD (`outputs/finish_20261023/oof_candidate_bootstrap.csv`).

SI-F.2 The elite subset and its permutation nulls

The confound is undiluted on the release's elite subset (the curated ≥ 2.0 mmol/g set, **not** a model's own top-k): $H(T|C) = 0.000$ bit there. Observed single-topology fraction **100.0%** (coarse, M = 448) against a row-level permutation null mean of **21.4%** (sd 1.0%; 95% interval 19.64–23.44%); in 10,000 row-level permutations **no simulated value reached the observed 100.0%**, so the Monte Carlo p is bounded by the simulation precision ($\sim 1 \times 10^{-4}$) rather than being an exact zero (`outputs/null_elite_permutation_row.csv`). A composition-level permutation (`outputs/null_elite_permutation_composition.csv`) gives null mean 100.0% with zero variance, because it holds fixed the property "each composition maps to a single topology" — the specific topology label of a composition may change, but the single-topology statistic is 1 by construction; the row-level unit is therefore the correct test, and the composition-level result is reported only to make that choice auditable. At fine granularity the null mean is already 86.3%, because singleton compositions dominate and remain single-topology under permutation, so the fine-grained 100.0% carries far less information.

SI-G. Ten targets and identity information

SI-G.1 The ten-target panel (descriptive)

Table S16. Ten-target panel: split effect and descriptor-only R^2 under both splits (25,000 samples).

Target	R^2 (random)	R^2 (topology hold-out)	ΔR^2	R^2 (descriptors only, random)	R^2 (descriptors only, topology)
CO_2 uptake 0.15 bar / 298 K	0.704	0.370	0.335	0.672	0.377
CO_2 uptake 0.10 bar / 363 K	0.518	0.113	0.405	0.488	**−0.076**
CO_2 uptake 0.70 bar / 413 K	0.587	**0.017**	**0.571**	0.535	0.044
excess CO_2 uptake 0.15 bar	0.973	0.972	**0.001**	0.981	0.980
CO_2 uptake (CO_2/N_2 binary)	0.700	0.377	0.322	0.665	0.388
N_2 uptake (CO_2/N_2 binary)	0.770	0.394	0.376	0.737	0.491
working capacity (VSA)	0.713	0.388	0.325	0.679	0.404
working capacity (TSA)	0.726	0.454	0.272	0.698	0.466
CO_2 heat of adsorption	0.861	0.744	0.117	0.848	0.755
$\log_{10}(CO_2/N_2$ selectivity)	0.825	0.726	0.100	0.813	0.711

The 0.704 / 0.370 pair differs from the main-text baseline (0.7066 / 0.3660) by ≤ 0.004 because this panel is a separate run using 100 rather than 120 trees. The split effect is positive on 10/10 targets (median +0.324, Wilcoxon p = 0.0020), but the targets are algebraically dependent, so n = 10 is not ten independent units and the p-values are descriptive, not confirmatory.

SI-G.2 Identity-block availability

Under the original representation, the observed identity effects differ across grouping schemes and covary descriptively with the availability of non-grouped identifiers. This availability pattern is a property of the fold construction; it is not a general rule for the value of identity features or an identified mechanism of the topology penalty. *Source data:* `outputs/chemistry_seen_grouping.csv`, `outputs/multitarget_confound_{metal,linker_pair}.csv`.

SI-H. ARC-MOF sources and granularity

SI-H.1 Non-singleton and sampling controls

The coarse single-topology fraction is 99.3% (256,351 rows, 180,989 compositions), 97.3% among the 49,869 non-singleton compositions; the fine values are 99.7% and 98.2% (26,380 non-singletons). The occupied-cell differences for the entropy correction are 3,989 (coarse) and 863 (fine), giving 0.0112 bit (15.2%) and 0.0029 bit (19.8%) against H(T|C) = 0.0739 and 0.0147 bit. *Source data:* `outputs/singleton_audit.csv` (all four values, both singleton counts and both corrections); the marginals are also in `outputs/confound_arcmof.csv`.

Why the audit N is 256,351. The ARC-MOF release has 264,225 structures; 7,874 lack the metal/linker cluster keys and are excluded from the coarse audit (256,351 rows). The fine key additionally requires a functional-group key, which 49,682 structures lack, leaving 214,543 rows. Rows are therefore dropped by key availability, not by topology; the topology column used here is `Crystalnet`.

Cluster IDs are chemistry-level labels generated from RAC descriptors. The release distributes three chemistry-cluster tables (`mc-clusters.csv`, `flig-clusters.csv`, `func-clusters.csv`) and a separate topology table (`all_topology_lists.csv`); its data description states that each chemistry table assigns each MOF to a cluster and that negative IDs denote "not in any cluster (i.e., assumed unique)" (ref. 15). More importantly, the ARC-MOF SI describes the diversity clustering as UMAP + HDBSCAN on RAC descriptors for metal-centre, ligand, functional-group and geometry types, with `min_samples = 1`, minimum cluster size 2 and `epsilon = 0.01`; points assigned −1 were treated as unique, and clusters were validated by the within-cluster standard deviation of RACs (ref. 3, SI "Diversity Analysis", Tables S3–S4). Topology is not among the clustering input types. The cluster IDs are therefore chemistry-level clusters, not per-topology labels; in the archived data they recur across topologies: 495 of 2,450 metal-cluster IDs (20.2%), 16,866 of 125,230 linker IDs (13.5%) and 7,272 of 43,745 functional-group IDs (16.6%) occur in more than one topology. This rules out the circularity in which each observed ID would be used in only one topology. Because the clustering inputs are RAC descriptors rather than topology labels, the observed cross-topology recurrence is not a per-topology artefact. *Source data:* `outputs/arcmof_cluster_span.csv`, produced by `scripts/arcmof_cluster_span.py` from the archived ARC metadata.

SI-H.2 Source removal and the DB0 condition

Removing one source sub-library at a time bounds any hMOF overlap, **only if** the hMOF-derived rows are concentrated in the largest sub-library (DB0); removing DB0 discards 79.0% of the analysed rows, so the exercise is a stress test on a fifth of the data, not a worst case over all of it.

Table S17. Leave-one-source-out single-topology fraction, coarse and fine columns (the two granularities are separate; the "any other source" rows report the minimum and maximum over the remaining sub-libraries).

Removal	coarse	fine
none	99.26%	99.74%
DB0 removed	**98.57%**	**98.96%**
any single other source (minimum)	99.22% (DB5)	99.72% (DB1)
any single other source (maximum)	99.46% (DB13)	99.84% (DB13)

Source data: `outputs/confound_arcmof_excl_derived.csv`, `outputs/arcmof_overlap.csv`.

SI-H.3 Within-source null on the common support set

Table S18. Per-source single-topology fraction against its within-source topology-permutation null (common support set, coarse chemistry, 1,000 permutations; the degenerate single-topology source DB6 is excluded). "Excess" is observed − null mean.

source	N	M	topologies	observed	null mean	95% reference	excess (pp)	H(T)
DB0	169,489	121,479	178	99.74%	77.17%	77.06–77.28	+22.6	2.29
DB1	11,546	9,008	19	100.0%	95.10%	94.89–95.29	+4.9	1.93
DB5	12,496	9,651	11	99.95%	94.45%	94.12–94.78	+5.5	0.69
DB7	4,653	3,588	19	99.36%	85.84%	85.23–86.48	+13.5	2.02
DB12	2,411	1,352	163	98.45%	61.14%	60.50–61.76	+37.3	5.59
DB13	4,207	3,220	10	92.30%	81.58%	80.93–82.17	+10.7	2.31
DB15	965	753	35	85.39%	82.84%	82.07–83.67	+2.6	3.74

The observed single-topology fraction sits above its own within-source topology-permutation null in every non-degenerate source, with excesses from **+2.6** (DB15) to **+37.3** (DB12) percentage points (Figure S6b; the per-source exclusivity rates of Figure S6a show the same spread at both granularities). The two nulls used in this section address different hypotheses and are reported separately: (i) the within-source permutation holds composition keys and topology-label counts fixed and permutes topology labels across records (1,000 permutations, Figure S8c); (ii) a common-rate binomial parametric bootstrap (20,000 draws) tests between-source heterogeneity. The excess shows a moderate-to-strong positive point estimate against H(T) (Pearson $r = +0.67$, $p = 0.10$; Spearman $\rho = +0.32$, $p = 0.48$; $n = 7$ non-degenerate sources), but with seven points and one influential row (DB12, H(T) = 5.59 bit, +37.3 pp) this is weak evidence; removing DB12 flips the point estimate (Pearson $r = -0.08$, Spearman ρ −0.09), so the association is not established. The source-to-source spread is 3,554× larger than a one-common-rate binomial model predicts at the fine granularity (about 2,000× at the coarse); in the 20,000-draw bootstrap **no simulated statistic reached the observed value**, so the reported Monte Carlo p is bounded by the simulation precision ($\sim 5\times10^{-5}$) rather than being an exact zero. The within-source permutation is a label-shuffle reference, not a test of a database-construction mechanism. *Source data:* `outputs/common_support/null_topology_cardinality.csv`, `outputs/source_confound_granularity.csv`, `outputs/source_spread_test.csv`.

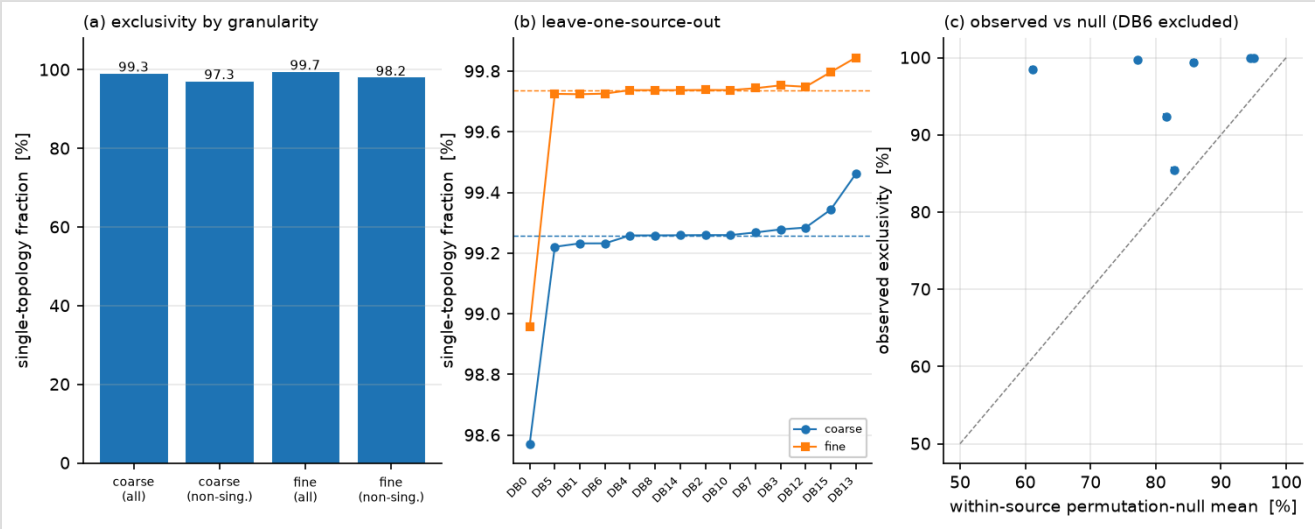


Figure S8. Bounded support audit in ARC-MOF (moved here from the main figures; the compact summary remains Table 3). **(a)** Single-topology composition fraction at four combinations (coarse/fine × all/non-singleton): 99.3, 97.3, 99.7 and 98.2. **(b)** Leave-one-source-out sensitivity: removing DB0 moves the fraction to 98.57–98.96, any other single source to 99.22–99.84. **(c)** Observed versus within-source topology-permutation null on the common support set, with the degenerate single-topology source (DB6) excluded; the permutation SD is below 0.5%, so the horizontal error bars are not visually distinguishable. *Source data:* `outputs/confound_arcmof.csv`, `outputs/confound_arcmof_excl_derived.csv`, `outputs/common_support/null_topology_cardinality.csv`.

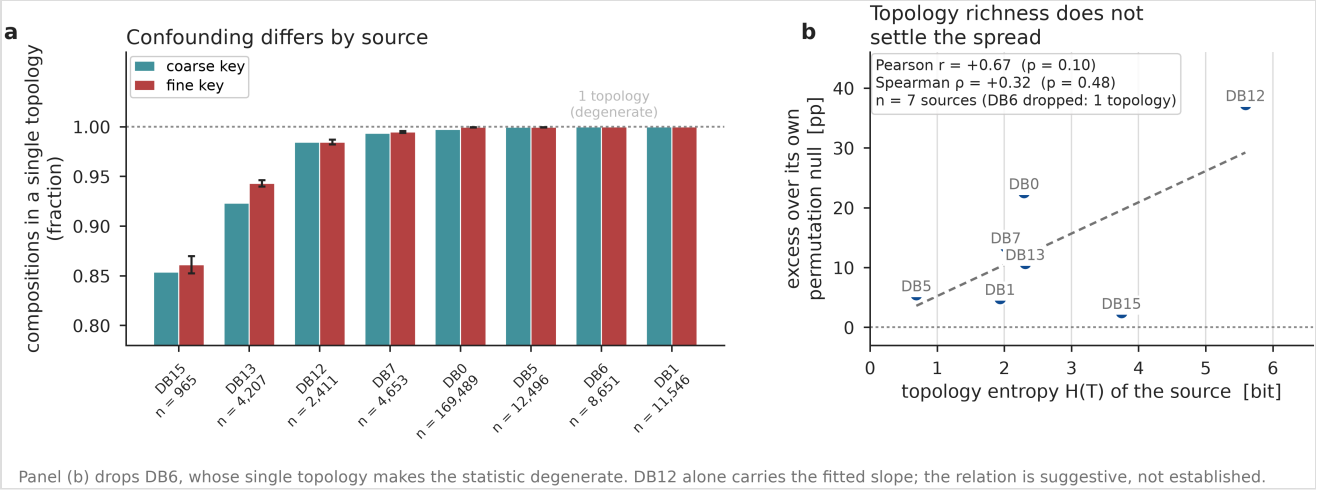


Figure S6. Confounding across ARC-MOF's source sub-libraries (n ≥ 200). (a) Single-topology chemistry fraction at coarse and fine granularity with bootstrap error bars (10 resamples); each sub-library's n is printed under its label. (b) Each source's confound degree above its own within-source permutation null (coarse, common support, B = 1,000) against H(T); the excess runs +2.6 pp (DB15) to +37.3 pp (DB12), DB6 excluded. The positive association is sensitive to DB12: Pearson r changes from +0.67 to −0.08 when that source is excluded. With seven non-degenerate sources, this does not establish a stable association. *Source data:* `outputs/source_confound_granularity.csv`, `outputs/common_support/null_topology_cardinality.csv`.

SI-I. Follow-up and sensitivity experiments: representation, support and screening limits

These follow-up analyses were integrated on 23 October 2026 and comprise exactly six experiments, in the order used in the abstract, main-text 2.7 and the cover letter: (1) representation (SI-I.1), (2) constituent novelty and numerical support (SI-I.2), (3) the CoRE structural-feasibility pilot (SI-I.3), (4) descriptive fold-level diagnostics (SI-I.4), (5) fixed-size partner replacement (SI-I.5) and (6) fixed-candidate-pool screening (SI-I.6). The ARC-MOF audit is not one of the six: it is the second-database support audit of main-text 3.5, with its controls in SI-H, and it adds no new performance or representation experiment. They complement the original analyses; they do not share a single replication unit, preprocessing rule or feature-encoding implementation. The new evidence is archived under `outputs/followup_20261023/` and belongs to the version-of-record deposit described in main-text 9 (v1.4), not to the earlier v1.3 release, which covers the original analyses only. No model was refitted during this manuscript-integration step. Tables S19–S24 can be recomputed from the archived metrics with `scripts/verify_followup_tables.py`. SI-I.7 adds a prespecified feature-subsampling sensitivity (Table S25) that *was* refitted on 24 October 2026 and is counted outside the six; its archive is `outputs/sensitivity_mf1/mapped/` and `outputs/sensitivity_mf1/full/`. That check establishes arithmetic consistency, not a fresh raw-data rerun.

SI-I.1 Representation comparisons (experiment 1)

Both targets were modelled: CO_2 uptake at 0.15 bar and 298 K (mmol g^{-1}), and $\log_{10}(CO_2/N_2$ selectivity), retaining only positive selectivity before transformation. G contains the eight numerical structure descriptors; these include mass and mass-normalized quantities and are not chemically orthogonal. B adds metal-node and functional-group identifiers. I adds the two linker identifiers in their original file slots. O adds topology to B; its training-fold encoding and feature-sampling configuration differ from the original 426-column analysis, so it is not an exact rerun of that model. T retains B and substitutes two parent-linker fingerprints for the linker-identifier blocks: 2,048 Morgan bits per slot, radius 2, no chirality. Constant fingerprint bits are removed using training rows only. Categorical encoders are fitted on training rows; unknown categories are ignored.

RF and ExtraTrees (ET) each use 120 trees and `max_features='sqrt'`. Five independent 25,000-row redraws use seeds 42, 143, 244, 345 and 446. The 100,000-row stage uses seed 42 only. Rows are sampled before removing non-finite values and negative numerical descriptors; no replacement rows are added. The targets therefore have different retained counts. The 100k run is a within-database sample-size check, not an independent or external test.

Parent-linker mappings were available for 33 of 58 identifiers, supported by automated graph-consistency checks; 21 have a second reference and 12 have a single reference. This is not expert confirmation of all framework-bound linker identities. The common mapped population retains 5,977/24,985 valid uptake rows in the seed-42 25k draw and 23,711/99,945 in the 100k draw. Only eight of eleven topologies remain; rht, tbo and the have no mapped rows. All mapped representations are compared on the same retained population. Fingerprint results are conditional on this mapping, parent representation, population selection and fixed model configuration.

Table S19. Paired pooled LOTO MAE changes. Each contrast is the first representation minus the second; negative values favour the first. The 25k columns show mean and sample SD across five paired redraws. The 100k column is one paired result. Uptake MAE is in mmol g^{-1}; log-selectivity MAE is on the $\log_{10}$ scale.

Population	Contrast	Target	Model	25k ΔMAE, mean ± SD	100k ΔMAE
Full	I-B	Uptake	RF	-0.008929 ± 0.005451	-0.005575
Full	I-B	Uptake	ET	-0.023178 ± 0.008336	-0.033010
Full	I-B	Log-selectivity	RF	-0.005480 ± 0.000609	-0.007276

Population	Contrast	Target	Model	25k ΔMAE, mean ± SD	100k ΔMAE
Full	I-B	Log-selectivity	ET	-0.006210 ± 0.002166	-0.007255
Mapped	T-I	Uptake	RF	+0.149485 ± 0.015124	+0.164800
Mapped	T-I	Uptake	ET	+0.160250 ± 0.014027	+0.153138
Mapped	T-I	Log-selectivity	RF	+0.039205 ± 0.001760	+0.042820
Mapped	T-I	Log-selectivity	ET	+0.054883 ± 0.003967	+0.055937

All five 25k contrasts are negative for full-population I−B and positive for mapped-population T−I, for both targets and ensembles. Pooled directions do not imply uniform per-topology effects or superiority over G. Resolving each contrast topology by topology (five-redraw mean of the paired difference) makes the reversals explicit. For uptake/RF, I−B is negative in 7 of 11 topologies and positive in bcu, fof, tbo and the, with mean differences from −0.023 to +0.035 mmol g^{-1}; 10 of 11 topologies keep the same sign in at least four of the five redraws. On the mapped subset, T−I is positive on average for uptake/RF yet negative in 3 of 8 topologies, and the largest single-topology change is +0.277 mmol g^{-1}. A pooled mean therefore summarises opposing local effects; the signs and counts for all four contrast–target–model combinations are in `outputs/finish_20261023/per_topology_direction_counts.csv`. For example, five-redraw uptake/RF mean MAE is 0.2510 for I and 0.2400 for G. Numerical inputs already contain chemical information, and dimensionality changes the feature-sampling opportunities of the trees. The experiment does not isolate a causal effect of chemical information. *Source data:* `experiment12/{full,mapped_common}/pooled_metrics.csv`, `experiment12/paper_tables/primary_LOTO_paired_directions.csv` and `experiment12/paper_tables/size_sensitivity_LOTO_paired.csv`, all beneath the follow-up archive.

SI-I.2 Constituent novelty and numerical support (experiment 2)

The primary coarse key is the ordered tuple (metal-node ID, linker-1 ID, linker-2 ID), without functional group. For each training/test split, CU denotes missing constituent identity; C0 has a seen coarse key; C1 has an unseen key but a seen metal-node ID and seen ordered linker pair; C2 has an unseen pair but a seen metal-node ID and both role-specific linker IDs seen separately; C3 has at least one unseen metal-node or role-specific linker ID. Sorted-pair and union-vocabulary alternatives are sensitivity analyses. The role distinction follows the source columns and does not establish a chemical site interpretation.

Numerical support uses nearest-neighbour Euclidean distance in the eight-descriptor space, standardized using training medians and IQRs; zero IQR falls back to training population SD and then one. A test row is outside at a distance above the training leave-one-out nearest-neighbour 95th percentile. This differs from the mean/SD definition in SI-I.4.

Table S20. Constituent novelty in the seed-42, full-population uptake I/RF LOTO audit. The classes partition all retained predictions in these two samples. MAE is in mmol g^{-1}. Class numbering is a support taxonomy, not an ordinal risk scale.

Class	25k n	25k MAE	100k n	100k MAE
C0	9,336	0.170034	38,490	0.178178
C1	4,624	0.268939	17,550	0.287387
C2	3,653	0.243445	14,441	0.262556
C3	7,372	0.349379	29,464	0.341106

Among unseen-key predictions, C1+C2 account for 52.9% at 25k and 52.1% at 100k. Thus exact-key loss often consists of recombining previously observed constituents. Approximately 96.9% of C3 rows involve unseen database metal-node identifiers; these are not unseen chemical elements. Linker-only unseen subclasses can be very sparse and do not justify general conclusions about new linker chemistry. In this specified I/RF audit, C1 MAE exceeds C2 at both sizes. At 25k, within each C0–C3 class, numerically outside rows have lower MAE than inside rows; target and topology composition can confound these comparisons. Neither pattern validates a monotonic novelty-to-error rule. *Source data:* `experiment12/full/novelty_error_metrics.csv`, `experiment12/full/C3_subtype_metrics.csv` and `experiment12/paper_tables/main_novelty_numerical_cross.csv`.

SI-I.3 CoRE structural-feasibility pilot (experiment 3)

The source is CoRE MOF 2019 ASR v1.1.0, Zenodo DOI `10.5281/zenodo.3370144`. Forty CIFs were randomly sampled with seed 20261022. The topology sub-pilot uses a preselected ten-structure subset. Counts below refer to successful parsing or returned technical objects, not identification accuracy.

Table S21. Completed technical steps in the CoRE pilot.

Pilot measure	Returned / completed
CIFs in source archive	12,020
Sampled CIFs parsed	40 / 40
Metal + organic candidates with bond cuts	38 / 40
Topology pilot: net key	9 / 10
Topology pilot: named-net match	6 / 10
Independent expert checks completed	0 / 40

The five independent-check fields (node, linker, periodic boundary, interpenetration, and source-publication checks) remain pending for all 40 records. A retry returning fragments does not resolve missing molecular identifiers. No database-wide composition support audit, comparable full-library topology annotation or external adsorption-prediction validation was completed. The zero completed expert checks must not be interpreted as zero correct structures. *Source data:* `experiment3/summary.json`, `experiment3/topology_pilot.json`, `experiment3/expert_review_sheet.csv` and `experiment3/sample_manifest.json`.

SI-I.4 Descriptive fold-level diagnostics (experiment 4)

This analysis attaches support summaries to the archived original LOTO fold MAE; it does not fit a risk model. The 66 fold records span six samples but reuse the same eleven topology groups. They are not 66 independent groups. Table S22 uses only the seed-42 100k sample. Its eight numerical descriptors are standardized by training means and population SDs (zero SD becomes one). The far threshold is the 95th percentile of leave-one-out nearest-neighbour distances for a deterministic sample of up to 5,000 training rows. Linker novelty uses the union of both training slot vocabularies. These metrics must not be identified with the role-specific classes or median/IQR distances of SI-I.2.

Table S22. Spearman correlations between topology-level diagnostics and absolute LOTO MAE. The second column uses eleven topologies; the last excludes the five-row rht group and retains ten topologies and 99,995 test rows. These are descriptive associations, without a claim of calibrated risk prediction.

Diagnostic	ρ, 11 topologies	ρ, 10 topologies (n ≥ 100)
Exact-key mismatch	+0.515932	+0.589077
Any building-unit novelty	+0.603023	+0.609723
Metal-node novelty	+0.575642	+0.547898
Linker novelty (union)	+0.193649	+0.189934
Mean numerical distance	-0.781818	-0.818182
Numerically far fraction	-0.763636	-0.818182
Novel and numerically far	+0.295573	+0.231363

Seven topologies have complete exact-key mismatch, yet their MAE ranges from 0.115300 to 1.260980 mmol g^{-1}. The original joint strata, cut at the eleven-group medians of unit novelty (0.1742241556) and far fraction (0.2340862423), contain 1/5/5/0 topologies in the low/low, low/high, high/low and high/high cells. These sparse strata do not establish complementary predictive value. Negative distance correlations do not imply that greater distance improves predictions. Absolute archived fold MAE is not a matched random-to-LOTO performance loss. *Source data:* `experiment4/fold_diagnostics.csv`, `experiment4/joint_group_strata.csv` and `experiment4/analysis_metadata.json`.

SI-I.5 Fixed-size partner replacement (experiment 5)

Within the seed-42 50k sample, the held-out test topology, nonpartner background, training size and partner fraction are fixed. Shared and nonshared partner rows are matched without replacement using outcome-blind greedy matching in the standardized eight-descriptor space, using the mean and population SD of the non-target-topology candidate pool (Euclidean caliper 0.5). The nbo direction uses 1,000 matched partner pairs and 4,000 background rows; pcu uses 726 pairs and 4,000 background rows. Partner doses are 0, 25, 50, 75 and 100% shared rows. These percentages describe the partner subset, not complete test coverage: even 100% gives only 78.75% and 15.08% exact-key coverage of the respective historical test sets.

Five seeds vary the model and replacement order, with the same fixed pools and test rows. At the endpoints, training membership is fixed across seeds. The RF has 120 trees and default `max_features=1.0`. Paired intervals below describe conditional learner variability, not uncertainty over new test sets, material populations or independent chemistry pairs. A separate paired sensitivity removes rows with negative numerical descriptors; only its two endpoints were refitted.

Table S23. Endpoint uptake MAE (mmol g^{-1}), averaged across five learner seeds. ΔMAE is 100% minus 0% shared partner rows. Intervals are paired t intervals (df = 4), conditional on the fixed pools.

Row policy	Test	MAE, 0%	MAE, 100%	ΔMAE	Paired 95% t interval
Historical rows	nbo	0.149276	0.158659	+0.009384	[+0.004902, +0.013865]
Historical rows	pcu	0.233135	0.239102	+0.005967	[+0.004025, +0.007909]
Negative descriptors removed	nbo	0.151084	0.158125	+0.007041	[+0.001748, +0.012335]
Negative descriptors removed	pcu	0.232815	0.238754	+0.005939	[+0.003533, +0.008346]

Both directions show increased endpoint MAE with shared partners in this local replacement design, including after descriptor cleaning. This does not demonstrate a causal disadvantage of shared chemistry. Residual differences remain in multivariate descriptor, metal-node, functional-group and target distributions: maximum metal total-variation distances are 0.095 and 0.171, and functional-group distances 0.340 and 0.416. Approximate numerical matching does not eliminate them. The cleaned test/training counts are nbo 7,663/4,991 and pcu 31,682/4,719. For the original deletion arms, the paired e−g interval includes zero on pcu (SI-C.4), so arm e's improvement over a does not establish a composition-specific advantage over same-count random partner deletion. *Source data:* `experiment5/curve_metrics.csv`, `experiment5/paired_endpoints.csv`, `experiment5/cleaned_endpoint_metrics.csv`, `experiment5/cleaned_endpoint_summary.csv` and `experiment5/matching_quality_summary.csv`.

SI-I.6 Fixed-candidate-pool retrospective screening (experiment 6)

One fitted model scores each entire frozen candidate pool, using the cleaned experiment-5 endpoint training sets. Candidate pools were examined in earlier analyses; this is a retrospective simulation, not an untouched external test. Categorical encoding is fitted on training rows only. G/B/I have the definitions above, and topology is not an input. RF and ET each have 120 trees with default `max_features=1.0`, unlike the `'sqrt'` setting in experiment 1. Absolute differences across these experiments cannot be attributed to representation alone. Five learner seeds keep training and candidate membership fixed.

Budgets are 1%, 5%, 10% and 20%, with k = ceil(N × fraction). True and predicted ties are resolved by the original source-row ID. Recovery is the overlap between predicted and oracle top-k sets divided by k. Regret is oracle mean uptake minus selected mean uptake; both refer to the same candidates. All G/B/I × RF/ET × 0/100%-shared arms and budgets are retained in the archived CSV.

Table S24. RF at the prespecified illustrative 5% budget: all twelve direction × partner-support × representation arms. Recovery SD reflects learner randomness only. Selected mean and regret are in mmol g^{-1}. The nbo candidate/training counts are 7,663/4,991; pcu counts are 31,682/4,719.

Test	Shared partners	Representation	k	Recovery (%), mean ± SD	Selected mean	Regret
nbo	0%	G	384	51.20 ± 0.93	0.433540	0.105741
nbo	0%	B	384	49.95 ± 0.56	0.429056	0.110225
nbo	0%	I	384	53.44 ± 0.56	0.437661	0.101621
nbo	100%	G	384	51.30 ± 0.88	0.431347	0.107934
nbo	100%	B	384	50.05 ± 0.65	0.428988	0.110294
nbo	100%	I	384	51.61 ± 1.16	0.433466	0.105815
pcu	0%	G	1585	20.08 ± 0.27	1.028581	1.012150
pcu	0%	B	1585	20.96 ± 0.24	1.041194	0.999537
pcu	0%	I	1585	15.81 ± 0.57	0.903649	1.137082
pcu	100%	G	1585	20.06 ± 0.42	1.029769	1.010962
pcu	100%	B	1585	21.35 ± 0.50	1.054243	0.986488
pcu	100%	I	1585	17.05 ± 0.62	0.933868	1.106863

The identity representation increases recovery relative to G for nbo but decreases it for pcu under nonshared partners. Within I, shared-partner replacement changes recovery in opposite directions across the two tests. Results also depend on budget and model. They establish local ranking sensitivity, not a universally preferable representation, reliable benefit of shared composition, prospective screening performance or a validated synthesis budget. *Source data:* `experiment6/screening_metrics.csv`, `experiment6/screening_summary.csv`, `experiment6/pool_audit.csv`, `experiment6/frozen_config.json` and `experiment6/candidates_*.csv`.

SI-I.7 Feature-subsampling sensitivity (`max_features=1.0`)

The representation comparisons above use `max_features='sqrt'`. Feature dimension changes how many descriptors a tree can sample at each split, so the same contrast was re-run with `max_features=1.0` and every other setting held fixed: 120 trees, the same sampling seed, the same targets and preprocessing, and the archived `split.npz` files reused row for row (no new sample, no new split). The setting was written into `frozen_config.json` before any fit, and all seeds, targets and folds are reported regardless of direction. Runs are archived under `outputs/sensitivity_mf1/mapped/` and `outputs/sensitivity_mf1/full/`; the analysis is `scripts/analyze_mf_sensitivity.py`, which asserts identical `raw_row_id` vectors on both sides before computing any difference.

Table S25. Mapped-subset I versus T contrast under the two feature-subsampling settings (5 seeds × 25,000-row draws; pooled leave-one-topology-out MAE in mmol g^{-1} for uptake and on the $\log_{10}$ scale for selectivity). "Gap" is T − I; "seeds > 0" counts the five redraws in which the gap is positive.

Target	Model	MAE I (sqrt → 1.0)	MAE T (sqrt → 1.0)	Gap (sqrt → 1.0)	Seeds > 0 (sqrt / 1.0)
Uptake	RF	0.3131 → 0.3144	0.4626 → 0.4365	+0.1495 → **+0.1221**	5/5 / 5/5
Uptake	ET	0.3269 → 0.3391	0.4872 → 0.4643	+0.1603 → **+0.1252**	5/5 / 5/5
Log-selectivity	RF	0.1443 → 0.1450	0.1835 → 0.1462	+0.0392 → **+0.0011**	5/5 / 3/5
Log-selectivity	ET	0.1488 → 0.1476	0.2036 → 0.1590	+0.0549 → **+0.0114**	5/5 / 5/5

Source data: `outputs/sensitivity_mf1/mapped/mf1_pooled.csv`, `mf1_paired.csv`.

Table S25b. Full-sample B versus I contrast under the two settings (5 seeds; RF; pooled LOTO MAE). "Gap" is I − B; "seeds < 0" counts redraws in which I is better.

Target	MAE B (sqrt → 1.0)	MAE I (sqrt → 1.0)	Gap (sqrt → 1.0)	Seeds < 0 (sqrt / 1.0)
Uptake	0.2599 → 0.2532	0.2510 → 0.2499	−0.0089 → **−0.0034**	5/5 / 5/5
Log-selectivity	0.1340 → 0.1382	0.1285 → 0.1321	−0.0055 → **−0.0061**	5/5 / 5/5

Source data: `outputs/sensitivity_mf1/full/mf1_pooled.csv`, `mf1_paired.csv`.

Three consequences. (i) For uptake, the T−I direction is unchanged: T is worse than I in all five redraws under both settings, and the gap is 18% (RF) and 22% (ET) smaller under the full-feature setting. (ii) For log-selectivity, the T−I gap nearly disappears under the full-feature setting (97% smaller for RF, where only 3 of 5 redraws remain positive; 79% for ET). (iii) Neither representation is uniformly better per topology: under the full-feature setting the uptake/RF gap is concentrated in pcu (+0.209) and etb (+0.132), with fof (−0.102) and nbo (−0.023) still favouring T, and pts changing sign from +0.010 to −0.038.

For the full-sample B/I contrast the direction is preserved in all five redraws for both targets: I remains better than B under the full-feature setting, by a smaller margin for uptake (−0.0034 against −0.0089) and by the same margin for log-selectivity (−0.0061 against −0.0055). The representation statements in the main text therefore hold under both settings, with the uptake magnitude explicitly configuration-dependent.

The two settings support different strengths of statement. "T did not improve pooled leave-one-topology-out MAE" holds for uptake and is the sentence used in the main text; for log-selectivity the correct statement is that the difference becomes negligible under the full-feature setting, not that fingerprints are neutral or harmful there. Random-split results are unchanged in direction (both representations improve slightly under 1.0), so the sensitivity bears on the topology-held-out contrast, not on the random-split level.

Limits. The full-feature tier covers the full 25,000-row samples for the B/I contrast with RF only: ExtraTrees requires about 150 s per fit at this setting, which exceeded the available compute, so no ET row exists for that tier (the mapped tier covers both learners). The mapped tier is 5,977 of 24,985 uptake rows and eight of eleven topologies, so it cannot represent the full sample. Neither tier is a hyperparameter search: one prespecified setting is compared with the archived one.
